

SymPlan: Enhancing Neurosymbolic Mathematical Reasoning in Compact Language Models via Representation and Planning

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Abstract. Mathematical reasoning with language models often relies on executable symbolic code. However, monolithic program synthesis requires simultaneously parsing problems, planning algebraic steps, and generating formal code, imposing severe cognitive load on compact small language models (SLMs, $\leq 8\text{B}$ parameters). Under 4-bit quantization, compact models frequently exhibit semantic misformulation, omitted domain boundaries, and silent regressions during code repair. We introduce **SymPlan**, an inspectable neurosymbolic framework designed for resource-constrained models. SymPlan structures problem solving into three stages: (1) a bounded *Fused Target-Anchoring Phase* (≤ 256 tokens) establishing an immutable target anchor with a formal output signature and algorithmic roadmap; (2) *Target-Anchored SymPy Code Generation* enforcing exact arithmetic and domain filters; and (3) *Targeted Debugging with Anti-Regression Selection*, pairing static AST diagnostics with lexicographic ranking to prevent valid solutions from being overwritten. Across six open-weight models (3B–8B) on MATH-500 under 4-bit quantization, SymPlan outperforms monolithic SymCode by +3.60% to +5.20% (averaging +4.20%) and Chain-of-Thought by +12.64%, while raising the execution success rate to 89.90% without fine-tuning.

Keywords: Neurosymbolic AI · Small Language Models · Mathematical Reasoning · SymPy Code Generation · Algorithmic Planning.

1 Introduction

Mathematical reasoning in language models has advanced through Chain-of-Thought (CoT) prompting [1] and structured search [2,3,4]. Nevertheless, autoregressive generation often produces arithmetic errors, omitted steps, and unfaithful derivations that resist verification [5,13]. Program-aided methods address these weaknesses by delegating computation to an external interpreter: PAL [6] and PoT [7] run numerical code in Python, MathCoder [8] and ToRA [9] blend tool calls with rationales, and SymCode [10] uses SymPy [11] for exact algebra.

However, successful program execution does not ensure mathematical correctness. Translation-based neurosymbolic frameworks fundamentally depend

on whether the generated formalization faithfully represents the problem [5,12]. Most existing approaches generate interpretation, planning, and code in a single prompt [6,10]. Although frontier proprietary models can often manage this combined task, compact small language models ($\leq 8\text{B}$ parameters) under 4-bit quantization [15] degrade severely on benchmarks such as MATH-500 [13,14]. In our experiments, these models display four consistent failure patterns: (1) *Semantic misformulation*: the generated program solves a distorted or incomplete equation; (2) *Target-anchor drift*: the program drifts from the queried objective and its type signature, such as returning an enumerated list when asked for set cardinality, or decimal floats instead of exact fractions; (3) *Domain boundary omission*: extraneous algebraic roots are accepted without filtering against domain constraints; and (4) *Silent repair regression*: iterative self-debugging yields syntactically valid scripts that replace previously correct answers with incorrect ones.

To address these issues, we present **SymPlan**, an inspectable neurosymbolic framework structured specifically for compact models. SymPlan divides problem solving into three stages: first, a *Fused Target-Anchoring Phase* establishes an explicit target anchor by extracting the queried variable, a formal output signature, and an algorithmic roadmap within a bounded step (≤ 256 tokens); second, a *Target-Anchored Code Generator* translates this specification into a self-contained SymPy program that enforces domain boundaries and exact arithmetic; third, a *Targeted Debugging Loop* couples runtime tracebacks and static AST diagnostics with an *Anti-Regression Selection* mechanism to preserve valid intermediate solutions.

Our primary contributions are: (1) characterizing formulation, target-anchor drift, and repair failure modes across six compact models on MATH-500; (2) introducing SymPlan, uniting fused target anchoring, specification-guided SymPy code generation, and anti-regression ranking; and (3) showing that SymPlan outperforms SymCode by +3.60% to +5.20% (and CoT by +12.64%) across six quantized SLMs, increasing ESR to 89.90% without fine-tuning.

2 Related Work

Program-Aided Reasoning. PAL [6] and PoT [7] offload numerical steps to Python, while MathCoder [8] and ToRA [9] combine natural-language derivations with code execution. SymCode [10] utilizes SymPy [11] for exact algebra. However, monolithic generation overburdens compact models: if the model misinterprets equations or omits boundary conditions, the interpreter returns mathematically invalid answers.

Formalization and Code Repair. Faithful CoT [5] separates reasoning from execution, and Logic-LM [12] refines symbolic formulations via solver feedback. For code repair, Self-Debugging [16] iteratively revises scripts via interpreter feedback. On compact models, unanchored repair frequently induces code cycling or silent regressions. SymPlan grounds the repair process: the immutable

target anchor preserves mathematical intent and output signature, while the attempt log prevents regressive overwrites.

Planning in SLMs. Structured prompting improves reasoning in compact architectures, including Self-consistency [2], Least-to-Most [3], and Plan-and-Solve [4]. SymPlan builds on these insights by establishing an explicit target anchor and structured execution steps designed for quantized models.

3 Method

3.1 Framework Formulation

Given a mathematical problem q , monolithic symbolic approaches [10] synthesize and execute code in a single unconstrained step: $C_{\text{base}} \sim \mathcal{M}(\cdot \mid q)$, $\hat{y} = \text{Exec}(C_{\text{base}})$. On compact models, this joint generation causes semantic drift and execution crashes. SymPlan structures problem solving through three target-anchored phases:

$$\mathcal{T} = (\tau, \Sigma, \mathcal{P}) = \text{FusedPlan}_{\mathcal{M}}(q), \quad (1)$$

$$C^{(0)} = \text{CodeGen}_{\mathcal{M}}(q, \mathcal{T}), \quad \mathcal{H} = \text{ExecuteAndRepair}_{\mathcal{M}}(C^{(0)}, q, \mathcal{T}), \quad (2)$$

$$\hat{y} = \text{SelectBest}(\mathcal{H}, \Sigma), \quad (3)$$

where τ denotes the queried target unknown, Σ is the formal output signature, $\mathcal{P}$ represents the ordered computational plan, and $\mathcal{H}$ records the execution history. Like a holding anchor stabilizing team structure, establishing $\mathcal{T}$ prior to code generation anchors mathematical intent throughout subsequent execution and debugging.

3.2 Fused Target Anchoring and Algorithmic Planning

Multi-turn prompting pipelines often double autoregressive latency and cause context drift in compact models. To avoid this overhead, SymPlan extracts the target anchor and plan tuple $\mathcal{T} = (\tau, \Sigma, \mathcal{P})$ in a single bounded generation (≤ 256 tokens): (1) **Target Specification (τ)**: the formal quantity queried, such as the positive real roots of an equation; (2) **Formal Output Signature (Σ)**: a discrete type signature $\Sigma \in \{\text{numeric}, \text{text}, \text{tuple}, \text{set}, \text{symbolic}\}$, where queries asking “how many” assign $\Sigma = \text{numeric}$ to return set cardinality rather than enumerating values; and (3) **Computational Plan ($\mathcal{P}$)**: an ordered sequence of symbolic operations $(p_1, \dots, p_m)$ defining symbols, equations, transformations, and domain boundary checks.

3.3 Target-Anchored SymPy Code Generation

Given the target anchor and plan $(\tau, \Sigma, \mathcal{P})$, generator $\text{CodeGen}_{\mathcal{M}}$ produces a self-contained SymPy script following four rules: (1) *Exact Arithmetic*: Rationals use `sp.Rational(p, q)` or `sp.S(p)/q`; float evaluations via `evalf()` are excluded

Table 1. Cross-model accuracy (%) and Execution Success Rate (ESR %) on MATH-500 across six compact models (3B–8B). Best accuracy is in **bold**.

Model	Size	Direct	CoT	SymCode	SymPlan	ESR (SC)	ESR (Ours)
Qwen2.5-3B-Instruct	3B	17.00	39.80	37.40	41.20	80.20%	87.20%
Qwen3-4B	4B	36.00	45.40	56.40	61.60	81.60%	90.20%
Qwen2.5-7B-Instruct	7B	25.40	45.60	54.00	57.60	88.40%	93.80%
Qwen2.5-Coder-7B-Instruct	7B	22.00	38.20	57.00	61.40	89.40%	92.40%
Llama-3.1-8B-Instruct	8B	23.20	37.60	37.60	41.80	76.60%	84.60%
Qwen3-8B	8B	36.20	45.00	59.80	63.80	86.80%	91.20%
Average (6 Models)	All	26.63	41.93	50.37	54.57	83.83%	89.90%

unless decimals are requested. (2) *Domain Filtering*: Roots from `sp.solve` are filtered against domain bounds ($x \in \mathbb{R}^+$, non-zero denominators). (3) *Simplification*: Expressions are simplified via `sp.together()` or `sp.cancel()`. (4) *Signature Conformance*: The script emits a single boxed $\text{\LaTeX}$ answer strictly conforming to signature Σ .

```
import sympy as sp; x = sp.Symbol('x', real=True); eq = sp.Eq(sp.sqrt(x), x - 2)
cands = sp.solve((x - 2)**2 - x, x); valid = [s for s in cands if s >= 2 and eq.lhs.subs(x,
s) == eq.rhs.subs(x, s)]
print(r"\boxed{%s}" % sp.latex(valid[0]))
```

3.4 Targeted Debugging with Anti-Regression Selection

Scripts execute in an isolated sandbox with timeout $\tau = 15\text{s}$ and retry limit $K = 2$, incorporating three safeguards: **AST Diagnostics**: Tracebacks $e^{(i)}$ are paired with static AST diagnostics (`lint_sympy_code`) and candidate verification $v^{(i)}$. **Anti-Cycling Repair**: If $C^{(i)} \equiv C^{(i-1)}$, an explicit constraint forces an alternative implementation strategy. **Anti-Regression Ranking**: Every attempt is logged into history $\mathcal{H} = \{r_0, \dots, r_k\}$ and scored via a lexicographic ranking tuple $\text{Rank}(r) = \langle \mathbb{I}_{\text{exec}}(r), \mathbb{I}_{\text{valid}}(r), \mathbb{I}_{\text{verif}}(r), -\text{attempt}(r) \rangle$, where $\mathbb{I}_{\text{exec}}$ indicates zero execution errors, $\mathbb{I}_{\text{valid}}$ denotes valid boxed extraction, $\mathbb{I}_{\text{verif}}$ indicates candidate verification pass, and $-\text{attempt}(r)$ prioritizes earlier valid runs over speculative later mutations. The optimal record is $r^* = \arg \max_{r \in \mathcal{H}} \text{Rank}(r)$. Multi-tier fallback handles unparseable outputs via regex scanning and type normalization.

4 Experiments and Results

Setup. We evaluate on the full MATH-500 benchmark [13,14] (7 subjects, 5 difficulty levels) across six open-weight models: Qwen2.5-3B-Instruct, Qwen3-4B, Qwen2.5-7B-Instruct, Qwen2.5-Coder-7B-Instruct, Llama-3.1-8B-Instruct, and Qwen3-8B. All models run under 4-bit NormalFloat4 (NF4) quantization [15] with greedy decoding (`temp = 0.0`, `max_tokens = 1024`). We benchmark against Direct zero-shot generation, Chain-of-Thought (CoT) [1], and monolithic SymCode [10], evaluating Accuracy (% EM) and Execution Success Rate (ESR).

Table 2. Fine-grained accuracy (%) across difficulty levels and subjects on MATH-500 for Qwen2.5-Coder-7B and Qwen3-8B.

Category	Count	Qwen2.5-Coder-7B		Qwen3-8B	
		SymCode	SymPlan	SymCode	SymPlan
<i>Difficulty Levels</i>					
Level 1–2	133	78.95	84.96	75.19	81.20
Level 3	105	61.90	73.33	63.81	66.67
Level 4–5	262	43.89	44.66	50.38	53.82
<i>Mathematical Subjects</i>					
Algebra	124	66.94	76.61	71.77	75.81
Counting & Probability	38	52.63	57.89	44.74	57.89
Geometry	41	31.71	34.15	46.34	51.22
Intermediate Algebra	97	51.55	48.45	50.52	51.55
Number Theory	62	66.13	70.97	62.90	72.58
Prealgebra	82	69.51	75.61	73.17	73.17
Precalculus	56	37.50	41.07	46.43	48.21

Table 3. Failure mode comparison between SymCode and SymPlan on MATH-500.

Failure Category	SymCode	SymPlan	Net Red.	Rel. Red.
Execution Failures (Syntax/Crash)	485 (16.2%)	303 (10.1%)	-182	-37.5%
Format / Type Mismatches	515 (17.2%)	394 (13.1%)	-121	-23.5%
Semantic Misformulations	974 (32.5%)	969 (32.3%)	-5	-0.5%
Total Incorrect Solutions	1489 (49.6%)	1363 (45.4%)	-126	-8.5%

Main Results. Table 1 reports results across the evaluated models. SymPlan consistently outperforms monolithic SymCode by +3.60% to +5.20% (averaging +4.20%) and CoT by +12.64% (reaching +23.20% on Qwen2.5-Coder-7B). Average ESR increases from 83.83% to 89.90% (+6.07%), lowering execution failures by 37.5%. In inference cost, SymPlan generates only 22.6% more tokens than SymCode (392.9 vs. 320.6 tokens/problem), remaining substantially shorter than CoT (423.7 tokens).

Stratified Analysis and Ablation. Table 2 indicates that gains concentrate in intermediate problems (Level 3: +11.43% on Coder-7B). By subject, improvements focus on *Algebra* (+9.67%), *Counting & Prob.* (+13.15%), and *Number Theory* (+9.68%), where integer constraints are easily omitted by monolithic baselines. In component ablations, target anchoring (+1.60% to +2.40%) and algorithmic planning (+2.40% to +3.80%) each improve accuracy independently, while combining them with anti-regression ranking achieves the highest accuracy (+3.60% to +5.20%).

Error Diagnostics and Case Studies. As detailed in Table 3, SymPlan reduces total errors by 126 across 3,000 evaluations (−8.5% relative drop). Safe SymPy templates reduce runtime crashes by 37.5%, and formal output signatures reduce format mismatches by 23.5%. On a degree-5 interpolation problem with $p(n) = n/(n^2 - 1)$ for $n \in \{2, \dots, 7\}$, monolithic SymCode produced runnable code yielding 39690 instead of 3/56 by omitting interpolation constraints; the

target anchor in SymPlan registered all six points, enabling exact Lagrange reconstruction. Conversely, semantic misformulations remained flat ($974 \rightarrow 969$, -0.5%), showing that neurosymbolic scaffolding resolves interface errors but cannot supply missing domain knowledge on complex non-constructive problems.

5 Conclusion

We presented SymPlan, an inspectable neurosymbolic framework for mathematical reasoning in compact models. By decoupling problem solving into fused target anchoring, specification-guided SymPy synthesis, and targeted debugging with anti-regression ranking, SymPlan resolves semantic misformulation, domain boundary omission, and repair regressions. Across six open-weight models (3B–8B) on MATH-500 under 4-bit quantization, SymPlan consistently outperforms SymCode by $+3.60\%$ to $+5.20\%$ (up to $+23.20\%$ over CoT) and raises the execution success rate to 89.90% without parameter updates. Future work will extend SymPlan to geometric diagrams and interactive theorem proving.

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